

ERH-WEI (WINNIE) SHENG

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EDUCATION

National Tsing Hua University

Hsinchu, Taiwan

*B.S. in Chemical Engineering & Computer Science
(Interdisciplinary Program of Engineering)*

Sep 2022 - Dec 2026

Selected coursework: Physical Chemistry I&II, Quantum Mechanics I, Statistical Thermodynamics, Mathematical Methods for Physicists, Linear Algebra, Design and Analysis of Algorithms, Computational Chemistry, Computational Condensed Matter Physics

RESEARCH INTERESTS

My interests revolve around representation learning that respects the symmetries of nature. I aim to use machine learning approaches such as neural quantum states to advance accurate variational modeling of quantum many-body systems beyond the practical reach of conventional ab initio methods, including strongly correlated systems.

RESEARCH EXPERIENCE

SU(2)-Equivariant Spinor Message Passing for Molecular Magnetism

Undergraduate Research Fellow, Caltech

Feb 2026 - present

- Anima AI+Science Lab; Advised by Prof. Anima Anandkumar
- Proposed eSpinor, an SU(2)-equivariant graph neural network that propagates spinor features to predict molecular magnetic anisotropy arising from spin-orbit coupling.
- **Manuscript (first author):** Submitted to the NeurIPS 2026 SIMBIOCHEM workshop; preparing for submission to PRX Intelligence.

Equivariant GNNs for Vibrational Circular Dichroism (VCD) Spectroscopy

Undergraduate Research Fellow, NTHU

Dec 2025 - present

- Lab for Material and Molecular Design; Advised by Prof. Kun-Han Lin
- Developing an equivariant GNN to predict VCD spectra of organic crystals. The goal is to assign absolute configurations in larger periodic systems using limited AIMD data.

GPU Acceleration of the OrbitAll Foundation Model

Undergraduate Research Fellow, Caltech

Jul 2025 - Nov 2025

- Anima AI+Science Lab; Advised by Prof. Anima Anandkumar
- Migrated OrbitAll's equivariant tensor-product backend to NVIDIA cuEquivariance, improving GPU efficiency and scalability for quantum-chemistry training.
- **Code release:** To be released as open source upon publication of OrbitAll.

AutoFF: Automated Force-Field Parameterization

Undergraduate Researcher, NTHU

Jun 2024 - Dec 2024

- Lab for Material and Molecular Design; Advised by Prof. Kun-Han Lin
- Developed an automated OPLS-AA parameterization toolkit to streamline molecular-dynamics simulations of organic semiconductors.
- **Presentation:** Honorable Mention, oral presentation, 2024 TwICHe National Conference.

RESEARCH EXPERIENCE

High-Entropy Alloys for the Hydrogen Evolution Reaction

Undergraduate Researcher, NTHU

Nov 2023 - Oct 2024

- Lab for Material and Molecular Design; Advised by Prof. Kun-Han Lin
- Used density functional theory to optimize hydrogen adsorption in high-entropy alloys by combining metals with different hydrogen affinities.
- **Manuscript:** Under review at JACS.
- **Presentation:** Poster, 2024 TwICHE National Conference.

LLM Reasoning and Evaluation in Low-Resource Languages

LLM Research Intern (Part-time), National Center for High-Performance Computing (NCHC)

Oct 2024 - June 2026

- Building culturally grounded corpora to evaluate and align LLM reasoning, with a focus on gender bias in low-resource languages.
- Co-developed a platform for LLM evaluation, annotation, and fine-tuning data preparation.

SELECTED HONORS & FELLOWSHIPS

Summer Undergraduate Research Fellowship (SURF), Caltech

2025, 2026

2nd place worldwide, ISC Student Cluster Competition (on-site, Hamburg)

2025

Represented Taiwan; qualified by winning the 2024 APAC HPC-AI Competition.

1st of 21 teams, APAC HPC-AI Competition (Asia)

2024

2nd of 9 teams, High Performance Application Competition (Taiwan)

2024

Grand Finalist (top 6), UAiTED Innovation Competition

2023

Selected from 300 participants across 16 partner universities in Asia.

2nd place & Taiwan representative, Taiwan International Science Fair

2021

3rd place & International Highlights, MOSTRATEC International Science Fair

2021

SKILLS

Programming: Python, C/C++, Julia, MATLAB, Fortran.

ML & HPC: Linux, PyTorch, CUDA, parallel programming (OpenMP, MPI), GPU performance profiling (NVIDIA Nsight), containers (Docker, Singularity), cluster management

Scientific software & libraries: VASP, AMS, GROMACS, LAMMPS, ASE, RDKit

Languages: English (full professional proficiency), Chinese (native)

Others: Git, $\text{\LaTeX}$, web & app development (HTML, CSS, JavaScript, Flutter)

LEADERSHIP & SERVICE

Captain, NTHU Student Cluster Competition Team

Led NTHU teams at national, Asia-Pacific, and international competitions.

Teaching Assistant

- Fundamental & Advanced High-Performance Computing Cluster Practice
- 2025 HPC $\times$ AI Winter Camp (Co-organized by NCHC & NTHU)

Conference Service

- Selected Student Volunteer, SC25 Conference
- Staff, QRACON (2024) and Chemistry National Meeting of Taiwan (2023)

Student Activities: protein simulations and data analysis for the NTHU iGEM team, marketing and visual design for the NYCU Niche team (STEM talks for girls), web development for the Student Society of Taiwan Advanced Rocketry (STAR)